

Technology Stack

Date	26 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component/ Layer	Technology Chosen	Justification/ Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Creates a simple, responsive, and user-friendly interface.
2	Backend / Server-Side	Python (Flask)	Handles business logic, loan eligibility processing, and connects the frontend with the ML model.
3	Machine Learning	Scikit-learn	Builds and predicts loan approval using trained ML models.
4	Database/ Data Storage	MySQL	Stores customer details, loan applications, and prediction history securely.
5	Data Processing	Pandas, NumPy	Cleans, processes, and analyzes loan data before prediction.
6	Development Environment	Visual Studio Code	Easy coding, debugging, and project management.
7	Version Control & CI/CD	GitHub	Stores source code and project documents with version tracking.
8	Operating System	Windows	Supports all development tools required for the project.
9	Cloud / Hosting/ Deployment	Render / Railway / PythonAnywhere	Hosts the web application online for users.
10	Third-Party Tools	Jupyter Notebook	Used for training, testing, and evaluating the ML model.